# Hints: Trigonometry – 2. Roots of unity and the complex exponential

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** For the filter: if $n \nmid m$ then $\varepsilon^m \neq 1$ and $\sum_{k=0}^{n-1}(\varepsilon^m)^k$ is a geometric sum equal to $\frac{1-\varepsilon^{mn}}{1-\varepsilon^m} = 0$; if $n \mid m$ every term is 1. For the subsets, encode them by the generating function $f(x) = \sum_s a_s x^s$, where $a_s$ is the number of subsets with sum $s$; show that $f(x) = \prod_{k=1}^{n}(1+x^k)$.

**Hint 2.** With $\omega = e^{2\pi i/3}$ the filter gives $\sum_{3|s} a_s = \frac{1}{3}(f(1) + f(\omega) + f(\omega^2))$. To compute $f(\omega) = \prod_k(1+\omega^k)$, group the factors in blocks of three consecutive $k$: $(1+\omega)(1+\omega^2) = 1$, so a full block contributes 2.

**Hint 3.** Writing $n = 3q + r$ one gets $f(\omega) = f(\omega^2) = 2^q$ for $r = 0, 2$ and $f(\omega) + f(\omega^2) = 2^q$ for $r = 1$; so the number of subsets (the empty one included) is $\frac{2^n+2^{q+1}}{3}$ if $3 \nmid n-1$ and $\frac{2^n+2^q}{3}$ if $n \equiv 1 \pmod 3$.

**Hint 4.** The same filter applied to $g(x) = (1+x)^n = \sum_k \binom{n}{k} x^k$ gives $\sum_{3|k}\binom{n}{k} = \frac{1}{3}(2^n + (1+\omega)^n + (1+\omega^2)^n)$; since $1+\omega = e^{i\pi/3}$, the answer is $\frac{2^n + 2\cos(\pi n/3)}{3}$.

## Problem 2

**Hint 1.** Divide $z^n - 1 = \prod_{k=0}^{n-1}(z-\varepsilon^k)$ by $z-1$: $1 + z + \cdots + z^{n-1} = \prod_{k=1}^{n-1}(z-\varepsilon^k)$. Put $z = 1$, take moduli, and use $|1 - e^{i\alpha}| = 2\sin\frac{\alpha}{2}$ for $0 < \alpha < 2\pi$.

**Hint 2.** For the sum, rewrite one term algebraically: for $|w| = 1$, $|1-w|^2 = (1-w)(1-\bar{w}) = -(1-w)^2/w$, hence $\frac{1}{|1-w|^2} = \frac{1}{1-w} - \frac{1}{(1-w)^2}$. So you need $\sum_k \frac{1}{1-\varepsilon^k}$ and $\sum_k \frac{1}{(1-\varepsilon^k)^2}$.

**Hint 3.** With $f(z) = \frac{z^n-1}{z-1} = \prod_{k=1}^{n-1}(z-\varepsilon^k)$ these sums are $(\log f)'(1)$ and $-(\log f)''(1)$. From $f(1) = n$, $f'(1) = \frac{n(n-1)}{2}$, $f''(1) = \frac{n(n-1)(n-2)}{3}$ one gets $\frac{n-1}{2}$ and $-\frac{(n-1)(n-5)}{12}$; add them.

## Problem 3

**Hint 1.** Write $\varepsilon = e^{2\pi i j/n}$ and $s = \varepsilon + \varepsilon^{-1} = 2\cos\frac{2\pi j}{n}$, a real number. Then $\varepsilon^2 = s\varepsilon - 1$: use this relation to reduce powers of $\varepsilon$.

**Hint 2.** Prove by induction that $\varepsilon^k = A_k(s)\varepsilon + B_k(s)$ with $A_k, B_k \in \mathbb{Z}[x]$ (take $A_{k+1} = xA_k + B_k$, $B_{k+1} = -A_k$). Hence $p(\varepsilon) = A(s)\varepsilon + B(s)$ with $A, B \in \mathbb{Z}[x]$.

**Hint 3.** If $\varepsilon$ is not real, compare imaginary parts: $A(s)\operatorname{Im}\varepsilon = 0$ forces $A(s) = 0$ and $z = B(s)$. If $\varepsilon = \pm 1$, $z = p(\pm 1) \in \mathbb{Z}$ is a constant polynomial.

**Hint 4.** Finally, $s = 2\cos\frac{2\pi j}{n} = D_j(2\cos\frac{2\pi}{n})$, where $D_0 = 2$, $D_1 = x$, $D_{k+1} = xD_k - D_{k-1}$ are integer polynomials with $D_k(2\cos\theta) = 2\cos k\theta$. Take $q = B \circ D_j$.

## Problem 4

**Hint 1.** Work in $\mathbb{C}$ and follow the hint of the sheet: write the vertices as a vector $Z = (z_0, \ldots, z_{n-1})$ and express the apex of the isosceles triangle erected on $z_j z_{j+1}$ with apex angle $\varphi$ in terms of $z_j, z_{j+1}$. (Check that, since $n$ is odd, no angle equals $\pi$.)

**Hint 2.** The apex is $w_j = \frac{1+ic_k}{2} z_j + \frac{1-ic_k}{2} z_{j+1}$ with $c_k = \cot \frac{\pi k}{n}$ for $\varphi = \frac{2\pi k}{n}$; the same formula covers the inward case $\frac{2\pi k}{n} > \pi$ (then $c_k < 0$). So the step is the linear map $A_k = \frac{1+ic_k}{2} E + \frac{1-ic_k}{2} S$, where $S$ is the cyclic shift, and the $(n-1)$-st polygon is $A_{n-2} \cdots A_1 Z$.

**Hint 3.** All $A_k$ are diagonal in the basis $v_m = (1, \varepsilon^m, \ldots, \varepsilon^{(n-1)m})$, $m = 0, \ldots, n-1$ (eigenvectors of $S$). Compute the eigenvalue $\lambda_{k,m} = \frac{1+ic_k}{2} + \frac{1-ic_k}{2} \varepsilon^m$ and show that it vanishes exactly when $m \equiv -k \pmod{n}$.

**Hint 4.** As $k$ runs over $1, \ldots, n-2$, $-k$ runs over all residues $2, \ldots, n-1$; so in the expansion $Z = \sum a_m v_m$ every component with $m \geqslant 2$ is killed, while those with $m = 0, 1$ survive. Hence the final vertices are $a_0 + b\,\varepsilon^j$: a regular $n$-gon centred at the centroid.

## Problem 5

**Hint 1.** Since $\varepsilon$ is a primitive $m$-th root of unity, $\prod_{k=0}^{m-1} (x - \varepsilon^k) = x^m - 1$; read off from Vieta the elementary symmetric functions $\sigma_j$ of $1, \varepsilon, \ldots, \varepsilon^{m-1}$: $\sigma_1 = \cdots = \sigma_{m-1} = 0$, $\sigma_m = (-1)^{m-1}$.

**Hint 2.** Lower bound: rank is subadditive, and $\sum_k (E + \varepsilon^k A) = mE + \sigma_1 A = mE$.

**Hint 3.** Upper bound: the factors $E + \varepsilon^k A$ commute, so $\prod_k (E + \varepsilon^k A) = E + \sigma_1 A + \cdots + \sigma_m A^m = E + (-1)^{m-1} A^m = 0$. Apply Sylvester's inequality $\operatorname{rank}(XY) \geqslant \operatorname{rank} X + \operatorname{rank} Y - n$ repeatedly to this product of $m$ factors.